

Employee Management System

Project Documentation

1. Project Introduction

The Employee Management System is a web-based application developed to manage employee records efficiently. The system allows authorized users to add, view, update, delete, search, and export employee information.

The main goal of this application is to replace manual employee record management with a centralized digital system that improves accuracy, accessibility, and data management.

2. Project Objectives

The main objectives of the system are:

- Maintain employee details digitally.
- Reduce manual paperwork.
- Provide easy employee record management.
- Allow users to perform CRUD operations.
- Generate employee reports in CSV format.
- Provide a simple and user-friendly interface.
- Secure employee information through authentication.

3. Technologies Used

Backend Technologies

Technology	Description
Java	Programming language used for backend development
Spring Boot	Framework for building the application
Spring MVC	Handles web requests and controllers
Spring Data JPA	Simplifies database operations
Hibernate	ORM framework for database interaction

Frontend Technologies

Technology	Description
HTML	Structure of web pages
CSS	Styling and user interface design
Thymeleaf	Server-side template engine

Database

Database	Usage
H2 Database	Used during development and testing

MySQL	Used for production deployment
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Build Tool

Tool	Purpose
Maven	Dependency management and project build

Development Tools

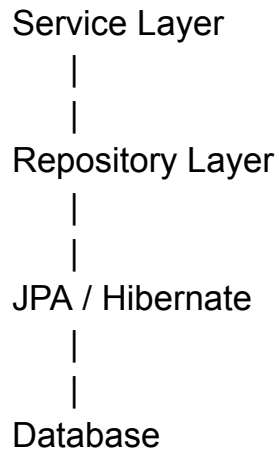
Tool	Purpose
Eclipse IDE	Application development
Browser	Application testing

4. System Architecture

The application follows a Layered MVC Architecture.

Architecture Flow

```
graph TD;
    A[User Browser] --> B[Thymeleaf View Layer];
    B --> C[Controller Layer];
    C --> D[ ];
```



5. Application Layers

Controller Layer

Responsibilities:

- Handles HTTP requests.
- Receives user input from the interface.
- Communicates between UI and business logic.
- Returns appropriate views.

Example:

EmployeeController.java

Service Layer

Responsibilities:

- Contains business logic.
 - Processes application operations.
 - Acts as a bridge between Controller and Repository.
-

Repository Layer

Responsibilities:

- Performs database operations.
- Uses Spring Data JPA interfaces.
- Handles CRUD operations.

Database Layer

Responsibilities:

- Stores employee information.
 - Maintains employee records.
 - Provides persistent data storage.
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6. Main Application Modules

Authentication Module

Features:

- User login.
- User logout.
- Session management.
- Secure page access.

Purpose:

Provides secure access to authorized users only.

Employee Management Module

Features:

Add Employee

Allows users to add new employee information.

Employee Fields:

- Employee ID
- Name
- Email
- Phone
- Department
- Designation
- Salary
- Joining Date
- Status

View Employees

Displays all employee records stored in the database.

Features:

- View employee details.
- Search employee records.
- Export employee data.

Update Employee

Allows modification of existing employee details.

Users can update:

- Personal information.
 - Department.
 - Salary.
 - Status.
-

Delete Employee

Removes employee records from the database.

Export Employee Data Module

Features:

- Export employee records into CSV format.
- Generate downloadable reports.

Exported information:

- Employee ID
 - Name
 - Email
 - Phone
 - Department
 - Designation
 - Salary
 - Joining Date
 - Status
-

7. Application Flow

Add Employee Flow

User enters employee details

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|

Submit Form

|

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Employee Controller

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Employee Service

|

|

Employee Repository

|

|

Database

8. Database Configuration

Application database configuration:

spring.datasource.url=jdbc:h2:mem:employee_db

spring.datasource.username=sa

spring.datasource.password=

spring.jpa.hibernate.ddl-auto=update

spring.jpa.show-sql=true

9. Database Entity

Employee Table

Column Name	Data Type
-------------	-----------

employee_id	Long
-------------	------

name	String
------	--------

email	String
-------	--------

phone	String
-------	--------

department	String
------------	--------

designation	String
-------------	--------

salary	Double
--------	--------

joining_date	Date
--------------	------

status	String
--------	--------

10. Security Implementation

The application uses Spring Security for authentication.

Features:

- User authentication.
- Protected application pages.
- Login session handling.
- Secure logout.

Example Login:

Username : admin

Password : admin123

11. Future Enhancements

The following features can be added in future:

- Role-based authentication.
- Advanced employee search.
- Pagination support.
- REST API integration.
- Cloud deployment.
- Email notifications.
- Advanced dashboard analytics.
- Employee attendance management.

12. Conclusion

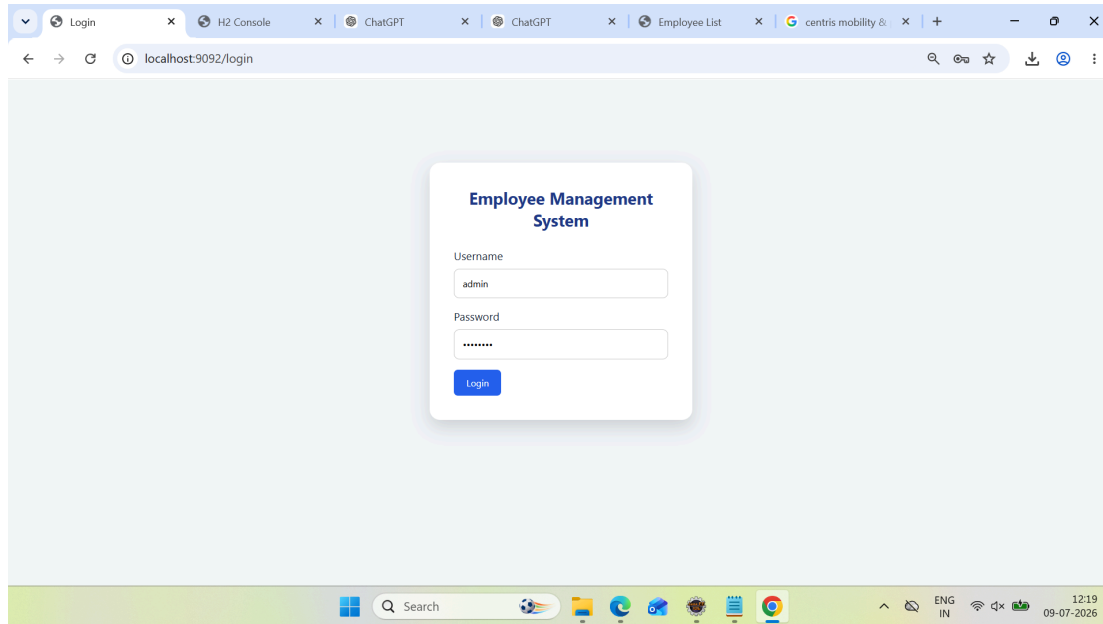
The Employee Management System provides an efficient solution for maintaining employee records digitally.

Using Spring Boot, Spring MVC, JPA, Hibernate, Thymeleaf, and database integration, the application provides a scalable and maintainable architecture.

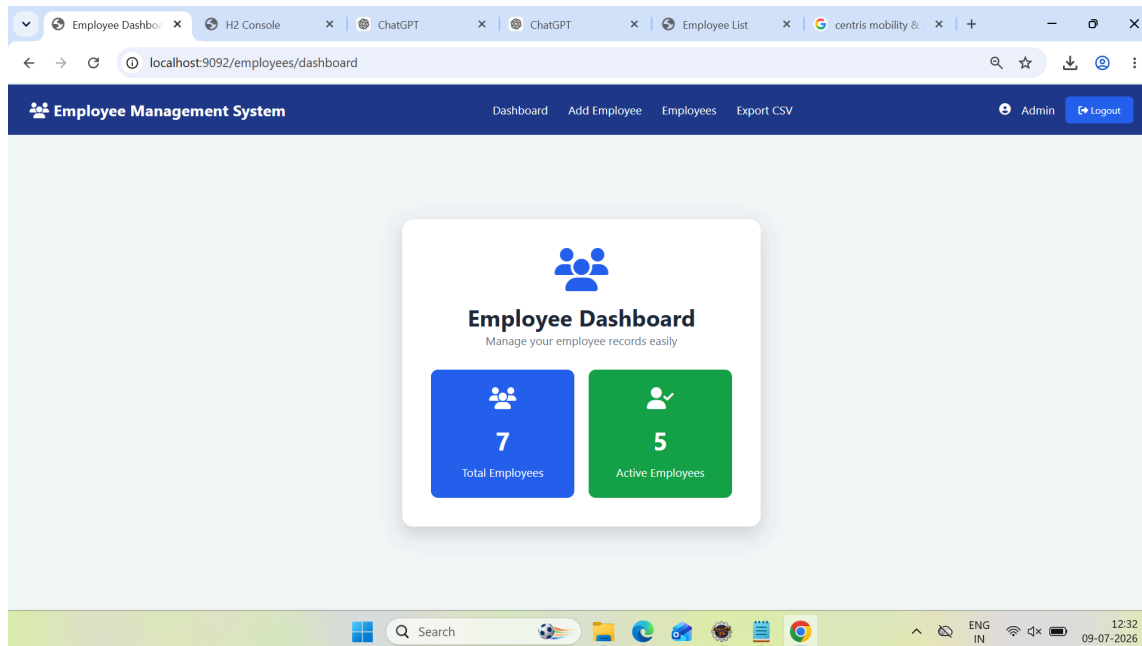
The system reduces manual work, improves data accuracy, and provides a foundation that can be extended for enterprise-level HR management solutions.

Screenshots

Screenshot 1: Login Page



Screenshot 2: Dashboard



Screenshot 3: Add Employee Page

Employee Management System | Dashboard | Add Employee | Employees | Export CSV | Admin | Logout

Add Employee

Name:

Email:

Phone:

Department:

Designation:

Salary:

Joining Date:

Status:

Screenshot 4: Employee List Page

Employee List

Search Name:

Search Department:

ID	Name	Email	Phone	Department	Designation	Salary	Date	Status	Action
1	vimal	vimal123@gmail.com	8015273600	IT	developer	50000.0	2026-07-10	Active	Edit Delete
2	kiran	kiran1234@gmail.com	8901324567	Testing	Unit Test	45000.0	2026-07-12	Inactive	Edit Delete
3	Raj	Raj@gmail.com	8765674323	IT	developer	47000.0	2026-07-14	Active	Edit Delete
4	Vijay	vijay123@gmail.com	9345564323	Network	Network Eng	51000.0	2026-07-21	Active	Edit Delete
5	Ajith	ajith123@gmail.com	7865341209	IT	software eng	64000.0	2026-07-01	Active	Edit Delete
6	Ashwin	ashwin123@gmail.com	8765899011	HR	HR	48000.0	2026-07-16	Active	Edit Delete

Screenshot 5: Update Employee Page

The screenshot shows a web browser window with the URL `localhost:9092/employees/edit/1`. The page is titled "Update Employee" and "Edit employee information". It contains a form with the following fields:

- Name: vimal
- Email: vimal123@gmail.com
- Phone: 8015273600
- Department: IT
- Designation: developer

The browser's taskbar at the bottom shows the Windows logo, a search bar, and several application icons. The system tray on the right indicates the language is "ENG IN", the date is "09-07-2026", and the time is "12:34".

Screenshot 6: H2 DataBase

The screenshot shows a web browser window with the URL `localhost:9092/h2-console/login.jsp?sessionId=52f9ed3a7a76bcee42b9b2a49455ab1f`. The page is titled "Login" and contains a form with the following fields:

- Saved Settings: Generic H2 (Embedded)
- Setting Name: Generic H2 (Embedded) [Save] [Remove]
- Driver Class: org.h2.Driver
- JDBC URL: jdbc:h2:mem:emsdb
- User Name: sa
- Password: []
- [Connect] [Test Connection]

The browser's taskbar at the bottom shows the Windows logo, a search bar, and several application icons. The system tray on the right indicates the language is "ENG IN", the date is "09-07-2026", and the time is "12:20".

Screenshot 7: Table Data's

Employee ListH2 ConsoleChatGPTChatGPTEmployee Listcentris mobility &+--oX

localhost:9092/h2-console/login.do?sessionId=52f9ed3a7a76bcee42b9b2a49455ab1f

Auto commitMax rows: 1000Auto complete: OffAuto select: On

jdbc:h2:mem:emsdbEMPLOYEEINFORMATION_SCHEMAUsersH2 2.4.240 (2025-09-22)

RunRun SelectedAuto completeClearSQL statement

select * from employee;

select * from employee;

EMPLOYEE_ID	DEPARTMENT	DESIGNATION	EMAIL	JOINING_DATE	NAME	PHONE	SALARY	STATUS
1	IT	developer	vimal123@gmail.com	2026-07-10	vimal	8015273600	50000.0	Active
2	Testing	Unit Test	kiran1234@gmail.com	2026-07-12	kiran	8901324567	45000.0	Inactive
3	IT	developer	Raj@gmail.com	2026-07-14	Raj	8765674323	47000.0	Active
4	Network	Network Eng	vijay123@gmail.com	2026-07-21	Vijay	9345564323	51000.0	Active
5	IT	software eng	ajith123@gmail.com	2026-07-01	Ajith	7865341209	64000.0	Active
6	HR	HR	ashwin123@gmail.com	2026-07-16	Ashwin	8765899011	48000.0	Active

(6 rows, 2 ms)

Edit

Windows Search

ENG IN

12:2009-07-2026